Rochester Institute of Technology

EEEE585/685 Tech Memo

From: Student A
Performed: September Xth, 2025
Due: September Xth, 2025
Subject: Lab #1 – Skeleton Guideline to Principles of Robotics Tech Memo Expectations

---

Abstract

General topics that need to be expressed in the abstract about the current robot of interest:

- *Configuration* → Differential drive, legged, arm, degrees of freedom, etc...
- *Locomotion* → Walking, rolling, prismatic, revolute, etc...
- *Programming* → Micro-controller, language, operating system, etc...
- *Control* → Controller boards, communication, extra hardware, etc...
- *Sensor(s)* → IR, ultrasonic, none, etc...
- *Behavior(s)* → Obstacle Detection/Avoidance, tele-operation, pre-determined sequences, etc...

Theory

Each of the experiments for the Principles of Robotics course is designed around a specific robot. Below is a list of suggested items with example questions that should be answered in the thoery section. The theory is not procedure, the procedure is provided in the lab handout, so:

- Motors
  - How/why is the motor typically used?
  - How does the particular motor operate?
  - What control signal is required?
  - How does the rotation/translation of the motor relate to the motion of the robot or its joints?

- Sensors
  - How/why is the sensor typically used?
  - What does the sensor measure?
  - How is it interfaced with the controller board(s) and/or programming language?
  - What control signals are required?
  - What signal is returned
- Calculations
  - What conversion equations were used?
  - What calibration processes were performed?
  - How were sensors, motors, etc. modeled to predict results?
  - Trial and error might actually be a suitable answer.
- High-Level Behavior
  - What is the behavior?
  - Why is the behavior used and in what context?
  - Does this behavior require feedback (what sensors?)
  - What actions are taken within the behavior?
- Locomotion
  - Why is this type of locomotion suitable for the robot?
  - What are the mechanics of the locomotion?

Results and Discussion

This section is twofold, one being the result and the other the discussion. The results are what you have accomplished in the lab using the provided theory and procedure in the handout. The discussion is explaining the results that you have seen or implemented. Below is a list of general results and corresponding discussion.

Results

- Custom locomotion or path planning, figures and tables might be useful in showing the results.
- Any specific equations used that were used that might expand on the theoretical equations explained in the theory.
- Calibration/conversion results.

- Tables of collected data from sensors.
- Charts plotting sensor data with correct trendline to represent the data and the equation of the trendline on the chart.
- Flow chart of behaviors.
- Screen captures of simulations.

Discussion

- Compare and contrast different locomotion schemes or path planning strategies .
- Explain the results of any equations or data collection, are they expected?
- Why was data collection performed? Was the data meaningful? How was the data used?
- Why was calibration or conversions needed?
- Explain all figures, tables, and equations.
- General Statement: Each lab requirement is there for a reason and therefore there can be something to be discussed about it.

Conclusions

A brief section containing one or two paragraphs that concisely states the nature/objective of the assignment, the approach taken to complete the assignment, and general observations about the outcome(s). Brief commentary on agreement - or lack thereof - between theory and experiment would be appropriate, but specific results that were already reported and discussed at length in the *Results and Discussion* **should not be reported here**.

Appendix

Include any relevant material that cannot be placed in the main body of the text without detracting from its appearance and readability.

- Include the questions.
- Large figures.
- Special code.
- References used, making sure to properly cite the reference throughout the memo.